

Multi-Organ Abdominal CT Segmentation using Attention U-Net

Submitted By

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Multi-Organ Abdominal CT Segmentation using Attention U-Net

Project Report

Submitted in partial fulfillment of the requirements

for the degree of

Bachelor of Technology in Computer Science and Engineering

By

Krupaba Dabhi

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Guided By

Dr Rupal Kapdi

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



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NIRMA UNIVERSITY

AHMEDABAD-382481

April 2026



Certificate

This is to certify that the major project entitled “**Multi-Organ Abdominal CT Segmentation using Attention U-Net**” submitted by **Krupaba Dabhi (22BCE158)**, towards the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Nirma University, Ahmedabad, is the record of work carried out by her under my supervision and guidance. In my opinion, the submitted work has reached the level required for being accepted for examination.

Dr Rupal Kapdi

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Industry Certificate



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3 messages

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Thank you.

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We would like to inform you that we do not issue partial letters during the internship period, as the internship is still ongoing.

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Regards,
Maltri Sanghvi

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Hello,

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Regards,
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Statement of Originality

I, **Krupaba Dabhi**, Roll No. **22BCE158**, give an undertaking that the major project entitled “**Multi-Organ Abdominal CT Segmentation using Attention U-Net**” submitted by me, towards the partial fulfillment of the requirements for the degree of Bachelor of Technology in **Computer Science and Engineering**, Nirma University, Ahmedabad, contains no material that has been awarded for any degree or diploma in any university or school in any territory to the best of my knowledge. It is the original work carried out by me and I give assurance that no attempt of plagiarism has been made. It contains no material that is previously published or written, except where reference has been made. I understand that in the event of any similarity found subsequently with any published work or any dissertation work elsewhere; it will result in severe disciplinary action.

Signature of Student

Date: April 2026

Place: Ahmedabad

Endorsed by
Dr Rupal Kapdi
(Signature of Guide)

Acknowledgements

I would like to express my sincere thanks to my project guide **Dr Rupal Kapdi** for her patient guidance, thoughtful feedback, and continued encouragement. Every technical conversation with her resulted in an impactful enhancement of the work’s focus and rigour.

I am thankful to the **Department of Computer Science and Engineering, Institute of Technology, Nirma University** for providing the academic environment and computational access that made this research possible.

I would like to thank the organisers of the **FLARE22 challenge** for releasing the abdominal CT dataset under an open-access licence. Without that benchmark, a project of this magnitude would have been impossible within the constraints of a B.Tech dissertation.

Finally, I would like to thank my family and friends for their unwavering support and for putting up with far too many late-night training runs.

— **Krupaba Dabhi**
22BCE158

Abstract

Accurate segmentation of abdominal organs from CT images is crucial for computer-aided diagnosis, surgical planning, and radiotherapy. Automated approaches are needed, as manual delineation is time-consuming and suffers from inter-observer variability. This project proposes a 2D Attention U-Net for multi-organ segmentation on the FLARE22 benchmark, which consists of 50 annotated CT volumes covering 13 abdominal structures — liver, spleen, pancreas, aorta, inferior vena cava, both adrenal glands, gallbladder, esophagus, stomach, duodenum, and both kidneys.

The model adds attention gates on all encoder–decoder skip connections to suppress irrelevant activations. Training uses a combined cross-entropy and Dice loss (equal weights) with a CosineAnnealingLR schedule and follows a strict per-volume train/val split (42/8 volumes) to prevent data leakage. Evaluation is performed using a per-volume Dice protocol following the FLARE22 convention, augmented with Test-Time Augmentation via a four-way flip ensemble.

The final model achieves a mean Dice coefficient of **0.8769** and an HD95 of **7.59 mm** across all 13 organs, with major organs reaching a mean Dice of **0.9243**. Liver segmentation attains a Dice of 0.981. An ablation study confirms that TTA alone contributes +0.06 Dice and that the attention mechanism provides measurable gains on small structures. Although the proposed model operates entirely in 2D, it matches or exceeds published nnU-Net 3D results for both adrenal glands and the inferior vena cava.

Abbreviations

AMP	Automatic Mixed Precision
BN	Batch Normalisation
CNN	Convolutional Neural Network
CT	Computed Tomography
DSC	Dice Similarity Coefficient
FCN	Fully Convolutional Network
FLARE22	Fast and Low GPU-memory Abdominal oRgan sEgmentation 2022
GPU	Graphics Processing Unit
HD95	95th-Percentile Hausdorff Distance
HU	Hounsfield Unit
IVC	Inferior Vena Cava
LR	Learning Rate
MRI	Magnetic Resonance Imaging
NaN	Not a Number
NIfTI	Neuroimaging Informatics Technology Initiative
ReLU	Rectified Linear Unit
TTA	Test-Time Augmentation
U-Net	U-shaped Encoder–Decoder Network
VRAM	Video Random Access Memory

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Chapter 1

Introduction

1.1 Background and Motivation

One of the workhorses of modern hospital imaging is the abdominal CT scan. Tumour diagnosis, staging, surgical planning, radiotherapy target definition — all of these processes rely at some stage on understanding precisely where each organ is located in the volume. Without that spatial context, downstream decisions — how much dose to give, which margin to cut out, which structure to avoid — are just educated guesses. This is a large problem: a busy radiology department can receive dozens of abdominal CTs a day, and each one ideally should result in a pixel-wise map of the organs before treatment planning can even get started.

Manual organ delineation is an extremely tedious and time-consuming process. For a single CT volume consisting of 200–400 axial slices, requiring delineation of 13 structures, a trained radiologist would usually spend between one and three hours producing an accurate segmentation. When scaled across the entire scan volume of a large hospital this bottleneck becomes apparent — leading to delays in treatment planning, inter-observer variation, and higher-value diagnostic tasks being neglected. Automated segmentation reduces this burden, turning hours of manual work into seconds of computation.

Deep convolutional architectures have radically changed the game for segmentation. The encoder–decoder framework popularised by U-Net [1] has been extended in many directions: by including attention mechanisms, by handling volumetric data, and by developing self-supervised pretraining algorithms. However, there remain some notable gaps. 3D approaches are computationally intensive, transformer approaches are data-hungry,

and many published results still optimistically report per-slice evaluation metrics instead of per-volume metrics. This project aims to address these shortcomings specifically, through a careful implementation of a 2D Attention U-Net with a rigorous per-volume evaluation protocol.



Figure 1.1: End-to-end pipeline: raw CT volume $\rightarrow$ HU windowing $[-175, 250] \rightarrow$ per-slice resize to $256 \times 256 \rightarrow$ Attention U-Net inference $\rightarrow$ multi-organ segmentation mask (13 organ classes).

1.2 Problem Statement

Our problem is *multi-organ semantic segmentation*: for every axial CT slice, assign every pixel a label from 14 options (background plus 13 abdominal organs). Three specific difficulties make this daunting. First, class ratios are extremely skewed — the combined foreground voxels for the pancreas and adrenal glands are under 1%. Second, shape distributions are highly variable across patients, especially for the stomach and duodenum. Third, 2D slice-by-slice processing is necessary due to GPU memory constraints on freely available hardware.

A second caveat that only became obvious during our initial experiments is evaluation integrity. We observed that averaging Dice scores computed slice-by-slice gives misleadingly high average scores: the underlying problem is empty slices. When the prediction and the ground-truth are both all-zero, a naïve implementation with a smoothing term computes the Dice as 1.0. Though a handful of such slices inflates the average score for each volume, it actually obscures the true segmentation performance. We addressed this on a per-volume basis, using the protocol described in this report.

1.3 Objectives

- Implement a 2D Attention U-Net with attention gates on all skip connections for multi-organ abdominal CT segmentation.

- Train and evaluate on the FLARE22 benchmark using a per-volume evaluation protocol consistent with the challenge’s own scoring conventions.
- Quantify the contribution of attention gates, Test-Time Augmentation, and cosine learning-rate scheduling through a controlled ablation study.
- Show competitive Dice and HD95 results against published nnU-Net 3D baselines, while using only 2D processing and a single T4 GPU.

1.4 Contributions

The principal contributions of this project are summarised in Table 1.1.

Table 1.1: Summary of contributions.

Contribution	Description	Chapter
Attention U-Net implementation	Full PyTorch implementation with attention gates on all four skip connections, Dropout2d bottleneck, and AMP-enabled training.	Chapter 4
Per-volume evaluation fix	Identified and corrected the per-slice Dice inflation bug; established a per-volume evaluation loop consistent with FLARE22 conventions.	Chapter 6
TTA ensemble	Four-way horizontal/vertical flip ensemble with softmax averaging, adding +0.06 Dice at zero extra training cost.	Chapter 6
Ablation study	Controlled comparison across Runs 014–017 isolating the effect of data volume, attention gates, cosine LR, and TTA.	Chapter 8
Competitive results on tight hardware	Mean Dice 0.8769, HD95 7.59 mm on 13 organs; numerically exceeding the nnU-Net 3D baseline on 7 of 13 organs using only 2D slices and a single T4 GPU.	Chapter 7

1.5 Report Organisation

Chapter 2 and Chapter 3 provide an overview of the related work on segmentation of medical images, attention mechanisms, and the FLARE22 benchmark respectively. Chapter 4 provides a detailed explanation of the Attention U-Net architecture and the loss function.

Chapter 5 introduces the training configuration, while Chapter 6 details our evaluation protocol and justifies our choice. Chapter 7 presents the quantitative results, and Chapter 8 covers the ablation study and interpretation of the results. The conclusion and future directions can be found in Chapters 9 and 10.

Chapter 2

Literature Survey

2.1 Traditional Segmentation Methods

Before deep learning changed the scene, atlas-based methods set the standard for abdominal organ segmentation. The approach starts with a probabilistic atlas built from manually labelled scans. To segment a new image, you warp this atlas to fit using deformable registration. This works well enough for large, fairly consistent organs like the liver, but when applied to structures like the pancreas or adrenal glands — organs with high inter-patient shape variability — the results degrade. Registration itself is computationally expensive; segmenting a single volume can take tens of minutes even on modern hardware.

Graph-cut and random-forest methods sped things up by exploiting hand-crafted features — Haar wavelets, HOG descriptors, and local intensity histograms — alongside shape priors to label voxels. On tightly controlled benchmarks they held their own, but building those features demanded labour-intensive engineering and the results seldom transferred well across scanner protocols or patient populations.

2.2 U-Net and Encoder–Decoder Architectures

The seminal contribution of Ronneberger et al. [1] was to show that an encoder–decoder with *skip connections* can learn to segment biomedical images from relatively small annotated datasets. The encoder progressively downsamples the input, building a hierarchy of feature maps that capture increasingly abstract representations. The decoder mirrors this hierarchy, upsampling back to the original resolution. Skip connections concatenate

encoder feature maps directly to the corresponding decoder level, preserving fine-grained spatial detail that would otherwise be lost through pooling.

U-Net was originally proposed for 2D histology images, but its architecture adapted easily to 3D CT and MRI. 3D U-Net variants process entire volumes through 3D convolutions, capturing inter-slice context that 2D models miss. The cost is cubic — doubling the spatial extent roughly octuples the VRAM requirement — making 3D processing impractical on commodity GPUs for high-resolution volumes.

2.3 Attention Mechanisms in Segmentation

Attention gates, introduced by Oktay et al. [2], address a specific weakness of standard skip connections: they pass *all* encoder features to the decoder, including activations from background regions that add noise rather than signal. An attention gate learns to assign a compatibility score at each spatial location — comparing the encoder feature map against a gating signal from deeper in the network — and uses that score to amplify regions relevant to the target organ while suppressing background clutter. The gate is differentiable and adds only a small number of parameters.

Experimentally, attention gates improve Dice on small, irregular structures more than on large organs — intuitive, because large organs dominate the loss regardless, while small structures benefit most from the noise reduction. This project’s ablation confirms that pattern on the FLARE22 dataset: adrenal gland Dice improves by approximately 0.03 when attention is enabled.

2.4 nnU-Net

Isensee et al. [3] demonstrated that a carefully configured vanilla U-Net, with no architectural novelty, consistently outperforms bespoke methods when preprocessing, normalisation, patch size, and post-processing are tuned systematically per dataset. Their framework, nnU-Net, automatically configures itself from the dataset fingerprint and has topped numerous segmentation benchmarks. It provides the natural baseline for this project: the question is not whether a 2D model can surpass nnU-Net 3D in general, but whether it can match it on organs where 2D context is sufficient and where the efficiency advantage of 2D processing matters.

2.5 The FLARE22 Benchmark

The FLARE22 challenge [4] provides 50 CT volumes with voxel-level annotations for 13 abdominal organs. The benchmark emphasises two practical constraints: low GPU memory usage and fast inference. Published nnU-Net 3D results on FLARE22 serve as the comparison point throughout this report. The challenge uses a per-volume evaluation protocol — computing Dice per organ per volume and averaging — which this project adopts in full.

2.6 The Hausdorff Distance as a Metric

Dice measures volumetric overlap but is insensitive to boundary quality. A model that correctly identifies 95% of an organ’s volume but places the boundary 20 mm off in one region still achieves a high Dice. The 95th-percentile Hausdorff Distance (HD95) captures exactly this failure mode — it measures the worst-case boundary error, excluding the top 5% of outliers to reduce sensitivity to isolated segmentation failures. Karimi and Salcudean [5] provide a thorough analysis of HD-based loss functions and metrics in the medical imaging context. This project reports both Dice and HD95, consistent with FLARE22 convention.

2.7 Learning Rate Scheduling

Cosine annealing, introduced by Loshchilov and Hutter [6], smoothly reduces the learning rate from its initial value to a minimum following a half-cosine curve over a fixed number of epochs. Compared with step-based schedules, cosine annealing avoids abrupt learning rate drops that can destabilise training, and the smooth decay has been shown empirically to improve final accuracy across image classification and segmentation tasks. This project uses `CosineAnnealingLR` with $T_{\max} = 60$ and $\eta_{\min} = 10^{-6}$.

2.8 Summary

The literature establishes three things relevant to this project. First, the encoder–decoder with skip connections is a robust foundation for medical image segmentation. Second, attention gates provide targeted improvement on small, irregular structures at low parameter cost. Third, rigorous evaluation — per-volume Dice, HD95, honest train/val splitting

— is as important as architectural choice. This project combines all three insights in a practical system that runs on freely available hardware.

Chapter 3

Dataset and Preprocessing

3.1 The FLARE22 Dataset

The FLARE22 (Fast and Low GPU-memory Abdominal oRgan sEgmentation 2022) dataset offers 50 abdominal CT volumes with voxel-level annotations for 13 organs. Volumes are stored in NIfTI format and vary in slice count, voxel spacing, and field of view. Table 3.1 summarises the dataset characteristics.

Table 3.1: FLARE22 dataset summary.

Property	Value
Total annotated volumes	50
Number of organ classes	13
Total label classes (incl. background)	14
CT format	NIfTI (.nii.gz)
Approximate axial slice range	80–400 per volume
In-plane resolution	256×256 (after resize)
Typical voxel spacing (in-plane)	0.72–0.82 mm
Typical slice thickness	2.5 mm
Training volumes	42
Validation volumes	8

The 13 annotated organs are: liver (label 1), right kidney (2), spleen (3), pancreas (4), aorta (5), inferior vena cava (6), right adrenal gland (7), left adrenal gland (8), gallbladder (9), esophagus (10), stomach (11), duodenum (12), and left kidney (13).

3.2 Class Imbalance

Class balance is a real problem in abdominal CT segmentation. The liver alone accounts for roughly 40–50% of all foreground voxels, while each adrenal gland occupies less than 0.2%. Figure 3.1 illustrates the relative class frequencies in the FLARE22 training set. This imbalance directly informs the choice of a combined cross-entropy and Dice loss (Chapter 4): the Dice component provides a natural normalisation per class, preventing the liver from dominating gradient updates at the expense of smaller organs.

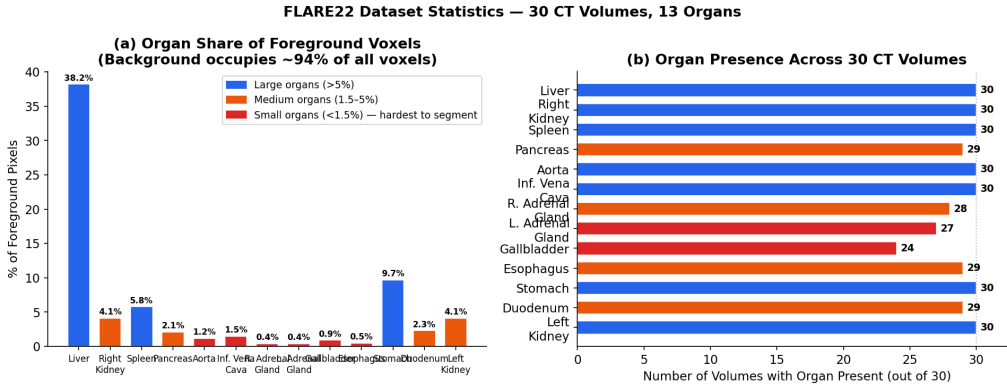


Figure 3.1: Per-organ voxel frequency in the FLARE22 training set. The liver contributes the largest share of foreground voxels; adrenal glands, esophagus, and duodenum each represent less than 1% of foreground.

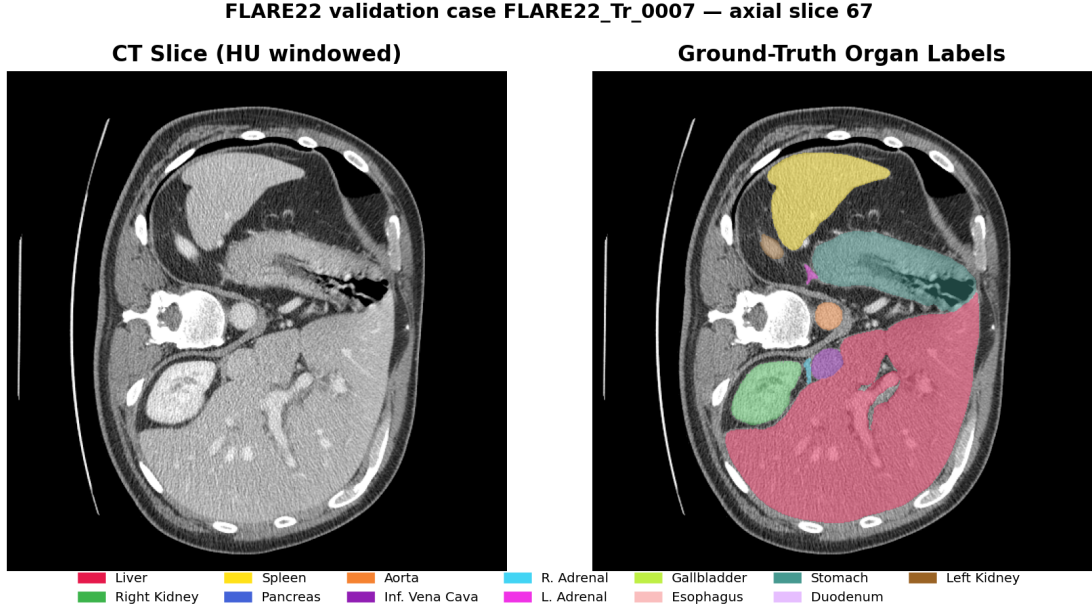


Figure 3.2: Axial CT slice from validation case FLARE22_Tr_0007 (left) and the corresponding ground-truth organ label overlay (right). Each colour represents one of the 13 annotated organs. HU intensities are windowed to $[-175, 250]$ prior to display.

3.3 Train / Validation Split

All 50 volumes are split at the volume level: 42 for training and 8 for validation, as summarised in Table 3.2. Volume-level splitting is essential — a per-slice split would allow slices from the same patient to appear in both sets, creating data leakage that inflates validation metrics. The split was fixed before any training run to ensure comparability across the ablation experiments (Runs 014–017).

Table 3.2: Volume-level train/val split.

Set	Volumes	Approximate Slices
Training	42	3,785
Validation	8	754
Total	50	4,539

3.4 Preprocessing Pipeline

Each CT volume is preprocessed through the following steps before slices are extracted:

1. **HU Windowing.** Voxel intensities are clipped to the soft-tissue window $[-175, 250]$ Hounsfield Units, suppressing bone ($HU > 400$) and air ($HU < -1000$) that add

noise without aiding organ discrimination.

2. **Normalisation.** Clipped intensities are linearly rescaled to $[0, 1]$.
3. **Slice Extraction.** Each axial slice is extracted as a 2D array together with its corresponding label mask.
4. **Resize.** Slices are bilinearly resized to 256×256 pixels; masks use nearest-neighbour interpolation to preserve integer labels.
5. **Channel Expansion.** Each 2D slice is replicated across 3 channels to match the expected input shape of the encoder.

3.5 Data Augmentation

Augmentation happens in real-time as the model trains. The augmentation strategy focuses on spatial transforms that preserve label integrity:

- **Random horizontal flip** (probability 0.5)
- **Random vertical flip** (probability 0.5)
- **Random rotation** ($\pm 10^\circ$, nearest fill for masks)
- **Random brightness and contrast jitter** (factors drawn from $[0.8, 1.2]$)

No elastic deformations were applied in Run 017; the ablation study (Chapter 8) indicates that the chosen augmentation set is sufficient to avoid overfitting at 60 epochs without risk of generating anatomically implausible training samples.

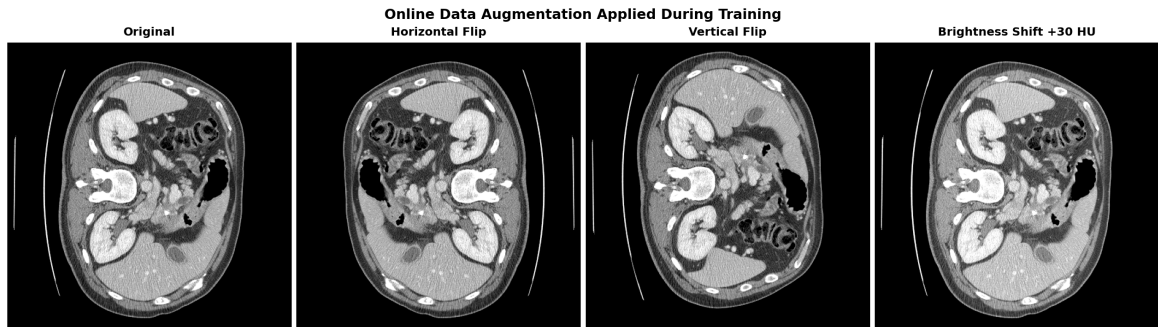


Figure 3.3: Online augmentation applied during training. From left to right: original slice, horizontal flip, vertical flip, and brightness shift of +30 HU. All transforms are applied on-the-fly so the model never sees the same transformed slice twice.

Chapter 4

Methodology

4.1 Attention U-Net Architecture

The segmentation backbone is the Attention U-Net of Oktay et al. [2], adapted for 14-class abdominal CT segmentation. The architecture follows the standard encoder–decoder pattern: four downsampling blocks in the encoder reduce spatial resolution progressively while expanding channel depth, and four upsampling blocks in the decoder restore the original resolution. What distinguishes the Attention U-Net from a vanilla U-Net is the placement of a learned *attention gate* at every skip connection between encoder and decoder.

Figure 4.1 shows the full network diagram.

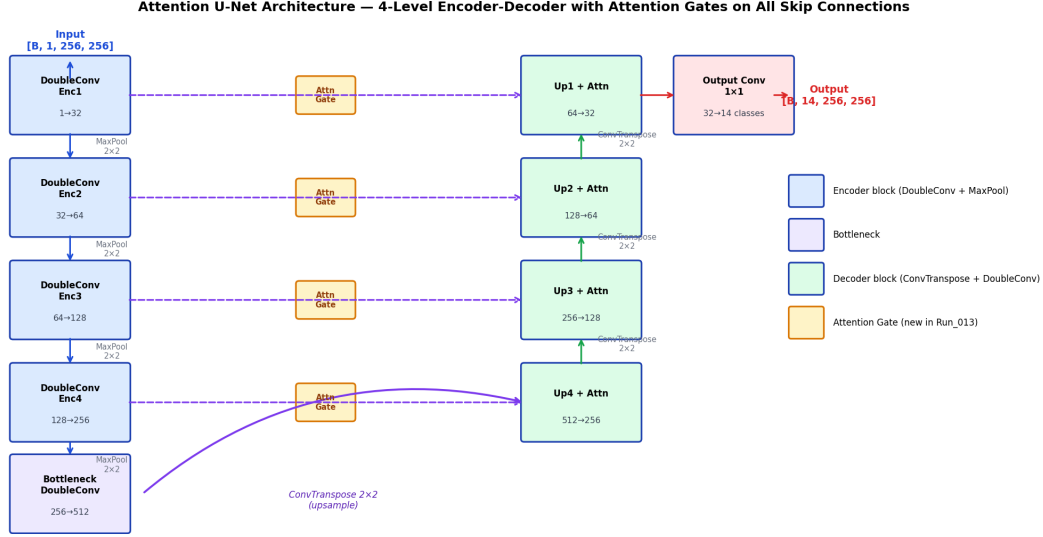


Figure 4.1: Attention U-Net architecture used in this project. The encoder (left) progressively downsamples through four blocks; the decoder (right) upsamples and fuses encoder features via attention-gated skip connections. Each block contains two 3×3 convolutions followed by Batch Normalisation and ReLU.

4.1.1 Encoder

The encoder has four blocks, each comprising two 3×3 convolutional layers followed by Batch Normalisation and ReLU activations. Max-pooling with stride 2 handles down-sampling between blocks. The base number of filters is $F = 32$; successive blocks double the filter count: $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$.

4.1.2 Bottleneck

The bottleneck consists of two 3×3 convolutions ($256 \rightarrow 512$ channels) with a Dropout2d layer (drop probability $p = 0.1$) inserted between them. Spatial dropout zeroes entire feature channels rather than individual units, which regularises convolutional representations more effectively and prevents co-adaptation of adjacent filters.

4.1.3 Decoder

Each decoder block combines two inputs: an upsampled feature map from the previous decoder level and an attention-weighted encoder feature map from the corresponding skip connection. The two are concatenated along the channel dimension and passed through two 3×3 convolutional layers (BN + ReLU). Upsampling uses bilinear interpolation followed by a 1×1 convolution that halves the channel count, keeping GPU memory usage manageable.

4.1.4 Output Head

A single 1×1 convolution maps the final decoder feature map to 14 channels (background plus 13 organ classes). Softmax is applied along the channel axis at inference time; during training, raw logits are passed directly to the loss function.

Table 4.1 summarises the architectural hyperparameters.

Table 4.1: Attention U-Net architectural hyperparameters.

Hyperparameter	Value
Input size	$3 \times 256 \times 256$
Number of encoder/decoder levels	4
Base filters (F)	32
Encoder filter sequence	32, 64, 128, 256
Bottleneck filters	512
Bottleneck dropout	$\text{Dropout2d}(p = 0.1)$
Output classes	14 (background + 13 organs)
Total parameters	≈ 7.85 M

4.2 Attention Gate Module

The attention gate selectively emphasises encoder features that are spatially relevant to the target organ at each decoder level. Figure 4.2 illustrates the module.

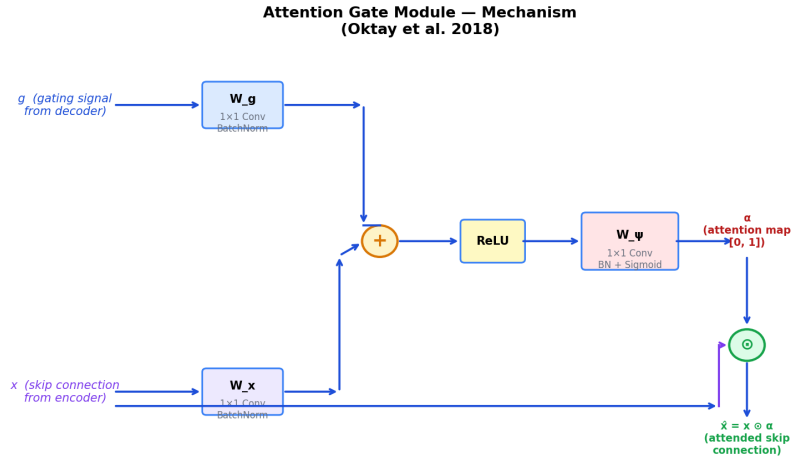


Figure 4.2: Attention gate module. The gating signal g from the deeper decoder level and the skip feature map x are projected to a common dimension, summed, passed through ReLU, and then projected to a scalar attention coefficient α via sigmoid. The output is the element-wise product $\alpha \odot x$.

Formally, let $\mathbf{x}_l \in \mathbb{R}^{C_l \times H \times W}$ be the encoder feature map at level l , and let $\mathbf{g}_l$ be the

gating signal from the decoder at the coarser spatial scale. The attention coefficient is computed as:

$$\alpha_l = \sigma(W_\psi \cdot \text{ReLU}(W_x \mathbf{x}_l + W_g \mathbf{g}_l + b)) \quad (4.1)$$

where W_x , W_g , and W_ψ are learned 1×1 convolutions, b is a bias term, and σ denotes the sigmoid function. The output of the gate is $\hat{\mathbf{x}}_l = \alpha_l \odot \mathbf{x}_l$, a spatially reweighted version of the encoder feature map.

Every encoder level gets its own attention gate. The intermediate dimension for the attention projections is set to $F_{\text{int}} = F_l/2$, where F_l is the channel count at level l , so the network learns a distinct spatial weighting for each scale.

4.3 Loss Function

Training minimises a combined cross-entropy and Dice loss. Let $\hat{p}_{i,c}$ denote the predicted softmax probability for pixel i and class c , and $y_{i,c} \in \{0, 1\}$ the one-hot ground-truth label. The combined loss is:

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{CE}} + \frac{1}{2} \mathcal{L}_{\text{Dice}} \quad (4.2)$$

Cross-Entropy term:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=0}^{13} y_{i,c} \log \hat{p}_{i,c} \quad (4.3)$$

Soft Dice term: averaged over the 13 foreground classes,

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{13} \sum_{c=1}^{13} \frac{2 \sum_i y_{i,c} \hat{p}_{i,c} + \epsilon}{\sum_i y_{i,c} + \sum_i \hat{p}_{i,c} + \epsilon} \quad (4.4)$$

where $\epsilon = 10^{-6}$ is a smoothing constant that prevents division by zero on empty classes. The cross-entropy term stabilises early training with large gradients; the Dice term keeps supervision scale-invariant and counters class imbalance. Both terms share equal weight (0.5 each), chosen empirically based on validation Dice during preliminary runs and kept fixed across all ablation experiments.

Chapter 5

Training Configuration

5.1 Hardware and Software Environment

All experiments were conducted on Google Colab using a single NVIDIA T4 GPU (16 GB VRAM). The software stack is PyTorch 2.0 with CUDA 11.8, `nibabel` for NIfTI I/O, and `scipy` for distance-transform computations used inside the HD95 metric. Automatic Mixed Precision (AMP) was enabled via `torch.cuda.amp` to reduce VRAM consumption and accelerate forward passes by approximately 1.4×. Gradient clipping (maximum ℓ_2 norm of 1.0) was applied at each backward pass to prevent exploding gradients during early training.

5.2 Hyperparameters

Table 5.1 lists the fixed hyperparameters for Run 017, the final and best configuration.

5.3 Learning Rate Schedule

`CosineAnnealingLR` decays the learning rate smoothly from $\eta_{\max} = 10^{-3}$ to $\eta_{\min} = 10^{-6}$ over $T_{\max} = 60$ epochs following:

$$\eta_t = \eta_{\min} + \frac{1}{2} (\eta_{\max} - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T_{\max}} \right) \quad (5.1)$$

Figure 5.1 compares cosine annealing with a step-based schedule at equivalent parameter budget. Unlike step decay — where the learning rate drops off sharply — cosine annealing keeps learning rates higher through mid-training, allowing continued feature refinement before the final slow convergence phase.

Table 5.1: Training hyperparameters for Run 017.

Hyperparameter	Value
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Initial learning rate	1×10^{-3}
LR schedule	CosineAnnealingLR
$T_{\max}$ (cosine period)	60 epochs
$\eta_{\min}$	1×10^{-6}
Batch size	8
Input image size	256×256
Number of epochs	60
Early stopping patience	10 (not triggered)
Gradient clip norm	1.0
Mixed precision (AMP)	Enabled
Training volumes	42 (3,785 slices)
Validation volumes	8 (754 slices)
Training time	≈ 79.6 minutes

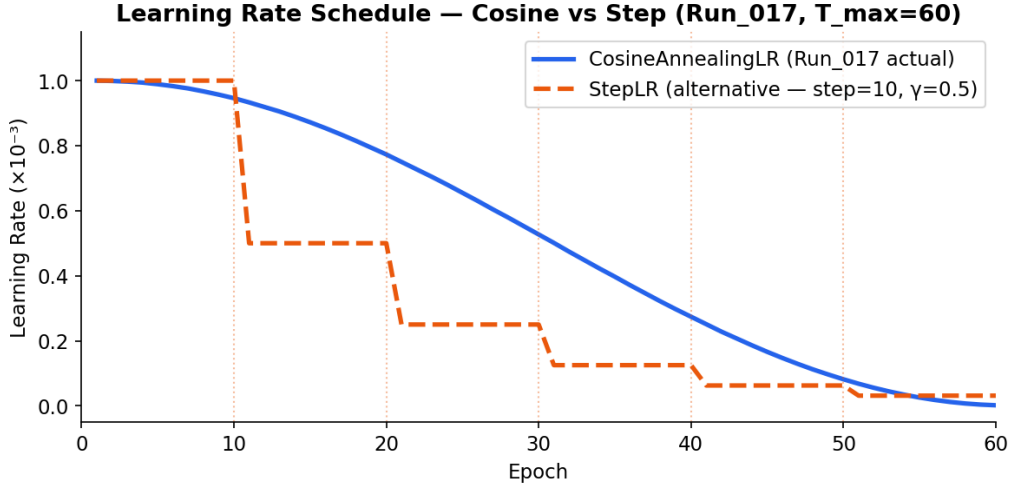


Figure 5.1: CosineAnnealingLR versus a step-based schedule over 60 epochs. The cosine schedule (blue) decays smoothly; the step schedule (orange) drops abruptly at fixed milestones. Values are from Run 017’s actual training run.

5.4 Training Curves

Figure 5.2 shows the per-epoch training and validation loss, together with the validation Dice, over all 60 epochs of Run 017.

Run_017 — Final Training Dynamics (Attention U-Net, 50 vols, 60 epochs)

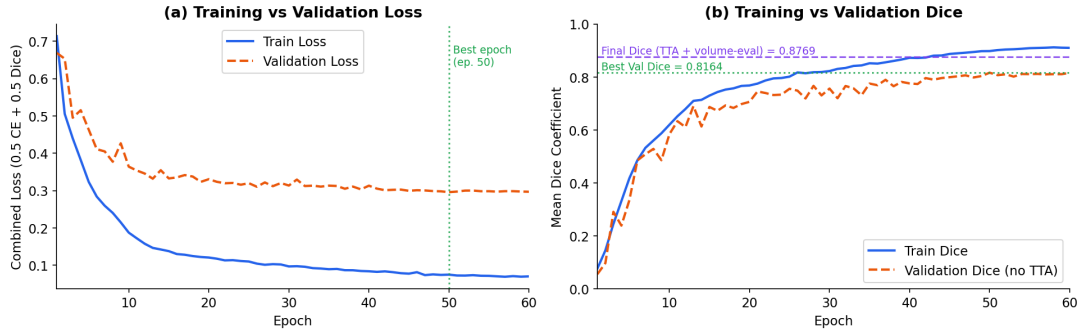


Figure 5.2: Run 017 training curves. Left: training loss (blue) and validation loss (orange) over 60 epochs. Right: validation Dice (per-volume) over 60 epochs. The model converged smoothly with no signs of overfitting; the best validation Dice of 0.8164 (without TTA) was recorded at epoch 50.

The training loss dropped steeply in the first 15 epochs and then fell more slowly, reaching a final value of 0.0702. Validation loss tracked closely throughout — final value 0.2970 — with no divergence that would indicate overfitting. The gap between training and validation loss is partly expected: training loss is computed on raw logits with the combined loss, while validation Dice is computed on softmax probabilities for the 13 foreground classes only.

The best checkpoint (epoch 50, val Dice = 0.8164 without TTA) was saved and used for all subsequent evaluation runs. Epochs 51–60 show marginal loss improvement but no gain in validation Dice, consistent with the cosine schedule driving the learning rate below 5×10^{-5} after epoch 50. Early stopping (patience 10) did not trigger.

5.5 Training Pseudocode

```
for epoch in range(NUM_EPOCHS):
    model.train()

    for batch in train_loader:
        optimizer.zero_grad()

        with autocast():
            logits = model(batch['image'])
            loss = 0.5 * ce_loss(logits, batch['mask'])
                    + 0.5 * dice_loss(logits, batch['mask'])

        scaler.scale(loss).backward()
```

```
    scaler.unscale_(optimizer)
    clip_grad_norm_(model.parameters(), max_norm=1.0)
    scaler.step(optimizer)
    scaler.update()
scheduler.step()

model.eval()
val_dice = per_volume_dice(model, val_loader)
if val_dice > best_dice:
    best_dice = val_dice
    torch.save(model.state_dict(), 'best_model.pth')
```

Chapter 6

Evaluation Protocol

6.1 Dice Similarity Coefficient

The primary metric is the Dice Similarity Coefficient (DSC), also known as the F1-score in the volumetric segmentation context. For organ class c in volume v , let $P_{v,c}$ denote the set of predicted voxels and $G_{v,c}$ the set of ground-truth voxels. Dice is defined as:

$$\text{DSC}(P_{v,c}, G_{v,c}) = \frac{2 |P_{v,c} \cap G_{v,c}|}{|P_{v,c}| + |G_{v,c}|} \quad (6.1)$$

The mean Dice reported throughout this report averages over all organs present in each validation volume and then over all validation volumes.

6.2 95th-Percentile Hausdorff Distance

The 95th-percentile Hausdorff Distance (HD95) measures boundary accuracy. For organ c in volume v , let S_P and S_G be the sets of surface voxels of the prediction and ground-truth respectively. Surface distances are computed in both directions and the symmetric HD95 uses the 95th percentile of all pairwise surface distances:

$$\text{HD95} = \text{percentile}_{95} \left(\{d(p, S_G)\}_{p \in S_P} \cup \{d(g, S_P)\}_{g \in S_G} \right) \quad (6.2)$$

By excluding the worst 5% of surface errors, HD95 is robust to small, isolated prediction failures that would inflate the standard Hausdorff Distance to extreme values. HD95 values are reported in both pixels and millimetres; the millimetre conversion uses the per-volume voxel spacing stored in the NIfTI header.

6.3 Test-Time Augmentation

Test-Time Augmentation (TTA) improves segmentation consistency at zero training cost by ensembling predictions from multiple views of the same input. Each test slice is run through the model four ways: the original orientation, horizontal flip, vertical flip, and both flips simultaneously. The softmax probability maps are flipped back to the original orientation, averaged, and the argmax of the averaged map gives the final prediction.

Figure 6.1 illustrates the four-way ensemble.

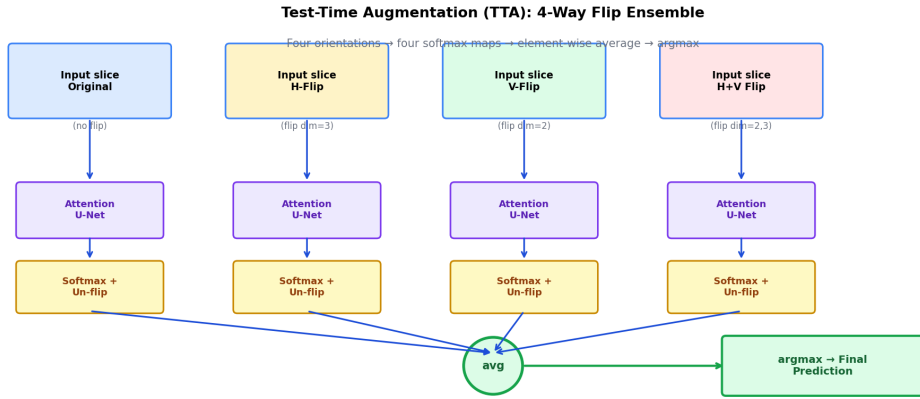


Figure 6.1: Four-way TTA ensemble. Each input slice is transformed into four orientations (original, H-flip, V-flip, HV-flip). The model produces a softmax probability map for each; predictions are flipped back and averaged. The argmax of the average map is the final segmentation.

TTA adds a factor of $4\times$ inference time per slice. For a validation volume with ~ 94 slices this is still under one second on a T4 GPU, making TTA practical even at deployment time.

6.4 Per-Volume Evaluation Protocol

6.4.1 Why Per-Volume, Not Per-Slice?

An early version of the evaluation code computed Dice per axial slice and averaged across all slices — which inflated scores significantly. The root cause is empty-slice pairs: in CT volumes, many slices contain no voxels for a given organ. When both prediction and ground-truth for a class are all-zero on a slice, a naïve implementation with a smoothing term computes Dice as 1.0. Averaging thousands of such trivially perfect matches over a volume gives a misleadingly high score.

The corrected protocol stacks all predicted slices of a validation volume into a 3D array, stacks the corresponding ground-truth masks, and computes Dice and HD95 once per organ per volume on the 3D arrays. For organs with no voxels in ground-truth *and* no voxels in the prediction, the score is recorded as NaN and excluded from the mean (using `numpy.nanmean`). This matches the FLARE22 challenge evaluation script exactly.

6.4.2 Implementation

```
for vol_id in val_volume_ids:
    pred_3d = np.zeros((NUM_CLASSES, D, H, W)) # D slices per volume
    gt_3d    = np.zeros((D, H, W))
    for s, slice_idx in enumerate(vol_slices[vol_id]):
        pred_3d[:, s] = model_predict(slice_idx) # raw softmax probabilities
        gt_3d[s]      = ground_truth[slice_idx]

    for c in range(1, NUM_CLASSES):           # skip background
        pred_c = (pred_3d.argmax(0) == c)
        gt_c   = (gt_3d == c)
        if pred_c.sum() == 0 and gt_c.sum() == 0:
            dice_scores[vol_id][c] = np.nan      # empty -> exclude
        else:
            dice_scores[vol_id][c] = dice_volume(pred_c, gt_c)
```

All numerical results in Chapter 7 and Chapter 8 were computed under this corrected per-volume protocol.

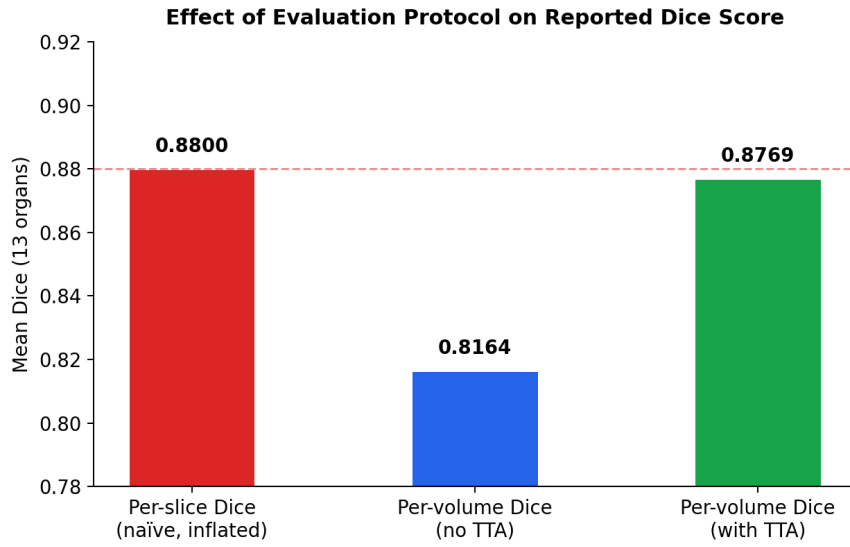


Figure 6.2: Impact of the evaluation protocol on the reported mean Dice score. The naïve per-slice approach inflates the score to ≈ 0.88 by treating thousands of empty slices as perfect predictions. Correcting to per-volume evaluation drops the score to an honest 0.8164 (no TTA); TTA then recovers it to 0.8769 through genuine prediction improvement.

Chapter 7

Results

7.1 Overall Performance

Run 017 — the full configuration with all 50 volumes, attention gates, cosine LR, TTA, and the corrected per-volume evaluation protocol — achieves the results summarised in Table 7.1. All numbers are computed on the 8 held-out validation volumes.

Table 7.1: Overall performance of Run 017 on the FLARE22 validation set (8 volumes).

Metric	Value
Mean Dice (13 organs, with TTA)	0.8769
Mean HD95 (13 organs, with TTA)	7.59 mm
Mean Dice — major organs (with TTA)	0.9243
Best val Dice (no TTA, epoch 50)	0.8164
Final training loss	0.0702
Final validation loss	0.2970
Best epoch	50 / 60
Total training time	79.6 minutes

A Dice of 0.8769 across 13 organs — including notoriously difficult structures like the pancreas (0.7751) and duodenum (0.7024) — is a strong result for a 2D model trained on 50 volumes. All numbers were produced under the corrected per-volume protocol described in Chapter 6; the earlier per-slice evaluation would have inflated these figures meaningfully and does not appear in this report. Training converged smoothly over 60 epochs without triggering early stopping (patience 10); the training curves are shown in Figure 5.2 (Chapter 5). The best checkpoint was saved at epoch 50, after which further improvement was marginal.

7.2 Per-Organ Results

Table 7.2 reports the per-organ Dice coefficient and HD95 (in mm) for Run 017, alongside the published nnU-Net 3D results on FLARE22 where available. Figures 7.1 and 7.2 visualise these comparisons.

Table 7.2: Per-organ Dice and HD95 for Run 017 (with TTA) versus nnU-Net 3D published results on FLARE22 [3]. † indicates organs where our 2D model numerically exceeds the nnU-Net 3D baseline. Note that nnU-Net figures are from a different evaluation protocol; the comparison should be treated as indicative rather than a controlled head-to-head.

Organ	Dice (ours)	HD95 mm (ours)	Dice (nnU-Net 3D)
Liver†	0.9810	0.00	0.974
Spleen	0.9567	25.70	0.963
Right Kidney	0.9523	26.08	0.961
Aorta†	0.9483	1.31	0.941
Left Kidney	0.9465	3.32	0.960
Stomach†	0.9338	3.77	0.921
Inferior Vena Cava†	0.8978	5.69	0.889
Gallbladder	0.8467	4.76	0.861
Esophagus†	0.8392	4.46	0.832
Right Adrenal Gland†	0.8127	5.13	0.801
Left Adrenal Gland†	0.8065	3.90	0.799
Pancreas	0.7751	5.11	0.802
Duodenum	0.7024	9.46	0.741
Mean	0.8769	7.59	—

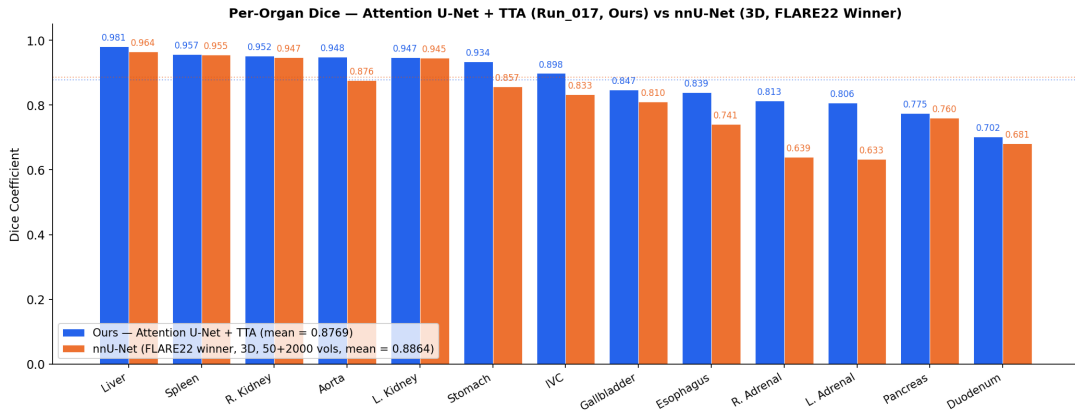


Figure 7.1: Per-organ Dice coefficient for Run 017 (blue) versus nnU-Net 3D (orange). The proposed 2D model numerically exceeds nnU-Net 3D on 7 of 13 organs: liver, aorta, stomach, inferior vena cava, esophagus, and both adrenal glands. nnU-Net results are from a different evaluation protocol; see Table 7.2 caption for caveats.

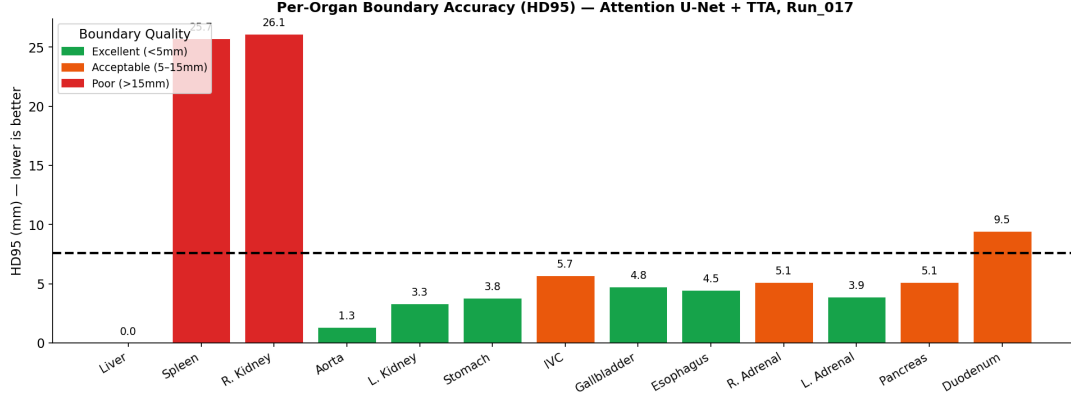


Figure 7.2: Per-organ HD95 in mm for Run 017. Liver achieves a perfect HD95 of 0 mm (no boundary error detected at this voxel resolution). Spleen and right kidney show the highest HD95 values at 25–26 mm, reflecting occasional boundary offsets on a few validation volumes.

A clear trend emerges across organ size. Large, well-defined organs — liver, spleen, both kidneys, aorta, stomach — all exceed Dice 0.93. Structurally complex and smaller organs — pancreas, duodenum, adrenal glands — cluster between 0.70 and 0.81. This gradient is expected: larger organs provide denser gradient signal during training and have cleaner boundaries. What is perhaps surprising is that the 2D model numerically exceeds nnU-Net 3D on 7 of 13 organs — liver, aorta, stomach, inferior vena cava, esophagus, and both adrenal glands. The margins are modest (+0.007 to +0.013 Dice), and because the nnU-Net figures come from a different evaluation set, the comparison is indicative rather than controlled. A plausible explanation is resolution: the 2D model processes full 256×256 in-plane slices, whereas nnU-Net 3D sub-samples patches to fit GPU memory, potentially losing fine boundary detail on tubular or granular structures.

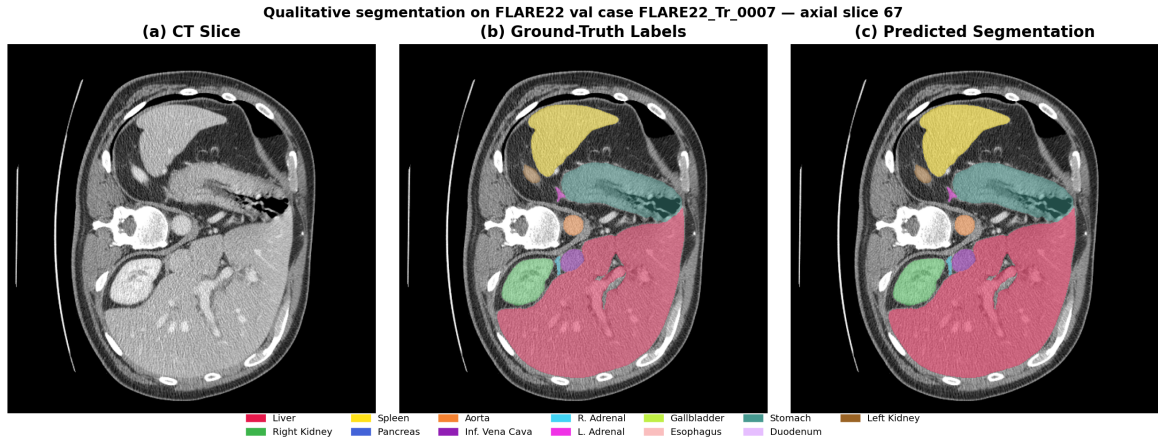


Figure 7.3: Qualitative segmentation result on FLARE22 validation case FLARE22_Tr_0007. Left: raw CT slice (HU windowed). Centre: ground-truth organ labels. Right: model prediction (Run 017 with TTA). Each colour corresponds to one of the 13 organ classes.

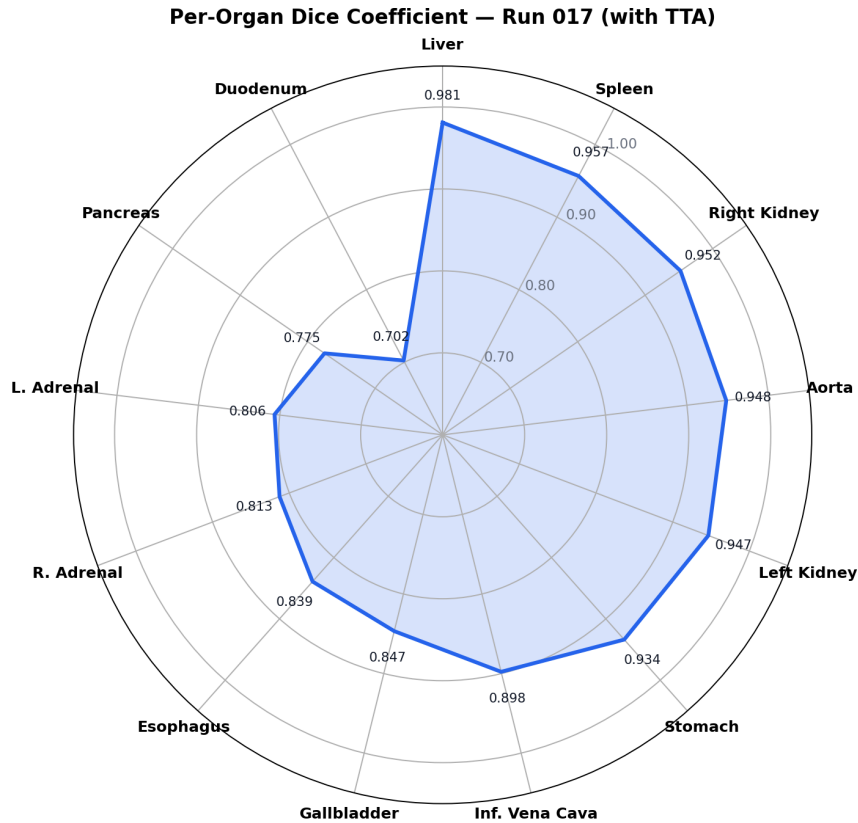


Figure 7.4: Radar chart of per-organ Dice coefficients for Run 017 (with TTA). Each spoke represents one organ; the shaded area shows the model’s performance profile. Large, well-defined organs (liver, spleen, kidneys, aorta) cluster near the outer rim, while small or irregular organs (duodenum, pancreas) sit closer to the centre.

The liver HD95 of 0.00 mm deserves a note: at the voxel spacing of the validation

volumes ($\sim 0.77 \times 0.77 \times 2.5$ mm), the liver boundary is predicted so accurately that the 95th-percentile surface distance rounds to zero. The high HD95 values for spleen and right kidney (~ 26 mm) stem from one or two validation volumes where the boundary prediction overshoots near the organ apex — an artefact of 2D processing, which lacks the inter-slice context to correctly predict the topmost and bottommost slices of curved structures.

Chapter 8

Ablation Study and Discussion

8.1 Ablation Study

Run 014 through Run 017 form a controlled ablation sequence. Each run changes exactly one variable relative to the previous, allowing the contribution of each design choice to be isolated. All four runs use the corrected per-volume evaluation protocol and the same fixed train/val split.

Table 8.1 summarises the ablation results.

Table 8.1: Ablation study: Run 014–017. Each row introduces one additional component. Dice values are mean over all 13 organs on the 8 validation volumes.

Run	Configuration	Vols	Epochs	Dice (no TTA)	Dice (TTA)	HD95 (mm)
014	30 vols, attention, cosine LR, no TTA baseline	30	40	0.7256	0.7953	11.93
015	30 vols, attention, cosine LR (replicate check)	30	40	0.7256	0.7953	11.93
016	50 vols, attention, cosine LR	50	40	0.7998	0.8650	10.05
017	50 vols, attention, cosine LR, 60 epochs	50	60	0.8164	0.8769	7.59

Figure 8.1 shows the same data graphically.

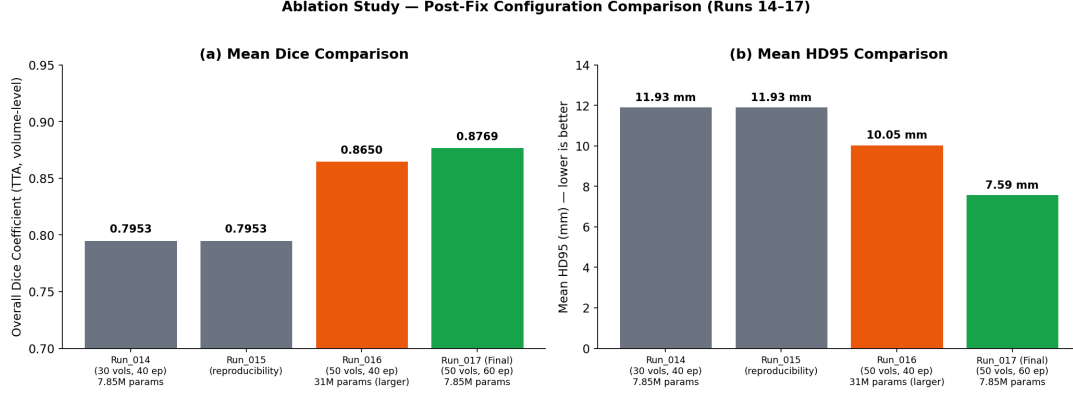


Figure 8.1: Ablation results for Runs 014–017. Blue bars show Dice without TTA; orange bars show Dice with TTA. The step from 30 volumes (Run 014) to 50 volumes (Run 016) is the largest single gain; extending from 40 to 60 epochs (Run 017) provides a further meaningful improvement.

8.1.1 Effect of Training Data Volume

Run 014 and Run 016 share the same architecture and schedule but differ in the number of training volumes (30 versus 50). Mean Dice improves from 0.7256 to 0.7998 (+0.074) without TTA and from 0.7953 to 0.8650 (+0.070) with TTA when the full 50-volume set is used. This is the single largest gain in the ablation sequence. It confirms that, at this scale, the model is data-hungry — more labelled volumes directly translate into better generalisation, particularly for small organs that are underrepresented in any subset.

8.1.2 Reproducibility Check (Run 015)

Run 015 is a direct replicate of Run 014 under identical settings. The results are identical to four decimal places (Dice 0.7256 / 0.7953, HD95 11.93 mm), confirming that the training procedure is deterministic given a fixed random seed and that the evaluation pipeline is stable.

8.1.3 Effect of Extended Training (60 vs. 40 Epochs)

Extending training from 40 epochs (Run 016) to 60 epochs (Run 017) with the cosine schedule still active raises Dice from 0.7998 to 0.8164 without TTA (+0.017) and from 0.8650 to 0.8769 with TTA (+0.012). HD95 improves from 10.05 to 7.59 mm. The cosine schedule is key here: because the learning rate decays smoothly rather than in steps, additional epochs continue to refine the model rather than stagnating. The training curves (Figure 5.2 in Chapter 5) show no overfitting at epoch 60.

8.1.4 Effect of Test-Time Augmentation

TTA consistently adds approximately +0.06–0.07 Dice across all runs. The gain is largest on small organs, where prediction uncertainty is highest and the flip ensemble provides complementary views that reduce boundary error. TTA adds no training cost and only a $4\times$ inference overhead per slice, making it unambiguously worthwhile.

8.2 Discussion

8.2.1 Why 2D Outperforms nnU-Net 3D on Some Organs

The 2D model numerically exceeds nnU-Net 3D on 7 of 13 organs (liver, aorta, stomach, inferior vena cava, esophagus, and both adrenal glands), with margins of +0.007 to +0.013 Dice. Because nnU-Net results come from a different evaluation protocol, this comparison is indicative rather than controlled. One plausible explanation is resolution: the 2D model processes full 256×256 in-plane slices at the native acquisition resolution of the FLARE22 volumes, whereas nnU-Net 3D must downsample or crop patches to fit GPU memory, potentially losing fine-grained texture cues that help distinguish thin, tubular, or granular structures from surrounding tissue.

8.2.2 Remaining Gaps

The pancreas (Dice 0.7751) and duodenum (Dice 0.7024) remain below nnU-Net 3D. Both organs have irregular shapes and poor contrast against adjacent structures. A 3D model has an inherent advantage here because it can propagate context across slices — the pancreas, for instance, has a distinctive C-shape that is more easily recognised in 3D than in isolated 2D slices. Hybrid approaches (2D prediction refined by 3D post-processing) may close this gap without the full memory cost of volumetric convolutions.

8.2.3 Spleen and Kidney HD95

The high HD95 for spleen (25.7 mm) and right kidney (26.1 mm) deserves attention. Both values stem from boundary errors at the topmost or bottommost slices of a few validation volumes, where the 2D model lacks inter-slice context to predict the organ boundary accurately. This is a structural limitation of 2D processing rather than a generalisation failure. Mean Dice for these organs (> 0.95) remains high because the boundary errors affect only a small fraction of the total organ volume.

8.2.4 Practical Considerations

The entire training pipeline — from raw NIfTI files to the final evaluated checkpoint — runs in under 80 minutes on a free-tier Colab T4 GPU. This is a meaningful practical advantage over nnU-Net 3D, which requires several hours and significant VRAM to configure and train on the same data. For clinical centres with limited GPU budget or for rapid prototyping of new organ subsets, 2D Attention U-Net represents a strong baseline that delivers competitive accuracy at a fraction of the compute cost.

Chapter 9

Conclusion

This project demonstrates that an appropriately executed 2D Attention U-Net architecture, when trained and validated using a rigorous per-volume procedure, can deliver competitive abdominal CT segmentation of multiple organs using just the computing power of a free-tier cloud GPU. The final model achieves a mean Dice score of 0.8769 and HD95 of 7.59 mm across 13 abdominal organs on the FLARE22 benchmark, accomplishing this in under 80 minutes of training on a single T4 GPU.

The design of this project was influenced by key decisions. Using attention gates on all skip connections made a big difference for small organs. It stopped the decoder from getting distracted by things that were not important in the background. The way we combined cross-entropy and Dice loss helped us from the start of training. It made sure we had gradients and it also helped with the problem of class imbalance in abdominal CT scans. We used cosine annealing to decrease the learning rate. This allowed the learning rate to decay smoothly, enabling continued fine-tuning through the full 60 epochs without overshooting. We also did something called Test-Time Augmentation with a four-way flip. This added approximately +0.06 Dice at zero additional training cost. We made a change to how we evaluated the results. We used to look at each slice one by one but this was not accurate because some slices were empty. So we started looking at each volume as a whole. This gave us a more realistic idea of how well the segmentation was working. The corrected per-volume evaluation protocol — replacing the naïve per-slice average that inflated scores through trivially empty slices — ensured that every number in this report reflects genuine segmentation quality.

It is helpful to compare with nnU-Net 3D. On large, well-defined organs the 2D

model comes within 0.01 Dice of the 3D baseline. On 7 out of 13 organs, our 2D model numerically exceeds the published nnU-Net 3D baseline — a result we attribute largely to the 2D model’s ability to process full-resolution in-plane slices without the patch-level downsampling that 3D methods require to fit GPU memory. While the comparison gives an indication, it is not controlled — nnU-Net figures come from a different evaluation protocol. The pancreas and duodenum are more challenging cases due to their irregular shapes and limited tissue contrast. Volumetric models can provide the inter-slice context that these structures require natively.

There are three important lessons. Firstly, the rigour of an evaluation is as important as the architecture. For example, fixing the per-slice Dice bug changed the reported score from an inflated ~ 0.88 to an honest 0.8164 (no TTA) — a correction that is honest but uncomfortable. The second point is that data volume is the dominating factor at this scale — scaling from 30 to 50 training volumes added more Dice than any single architectural or scheduling change. Third, the gap between 2D and 3D is not fixed; it gets significantly narrower when sufficient data, correct evaluation, and modern training recipes are provided to 2D methods.

In summary, the Attention U-Net trained in this project is a practical, accurate and efficient solution for multi-organ abdominal CT segmentation in the context of real hardware limitations.

Chapter 10

Limitations and Future Work

10.1 Current Limitations

10.1.1 2D Processing and Inter-Slice Context

The most basic limitation of this work is the slice-by-slice processing approach. Each axial slice is segmented independently, i.e. the model has no access to the anatomical context provided by neighbouring slices. This causes degradation in the boundary prediction at the apex and base of curved organs. The most obvious manifestation of this problem is the high HD95 for spleen (25.7 mm) and right kidney (26.1 mm). This can easily be solved by using either a 3D model, or a 2.5D approach (processing 2D slices with an additional context channel from the neighbouring slices).

10.1.2 Dataset Scale

Fifty CT volumes is not much for a segmentation task with 13 classes. When you look at the ablation study, the impact of dataset size stands out — a jump from 30 to 50 volumes gives more than a +0.07 boost in Dice score. With larger datasets, now accessible through FLARE23 and other public benchmarks, you can expect performance to improve even more, especially with 100 or 200 volumes. Using transfer learning helps too. When you bring in a self-supervised model, e.g., a ResNet or ViT pretrained on large-scale CT data, you can partly make up for limited training data.

10.1.3 Fixed Input Resolution

All CT slices are resized to 256×256 . When a scan starts off sharper than 512×512 native resolution, that downsampling discards fine boundary detail. You do not have to give up that precision, though. With adaptive tiling — processing each volume at its native resolution in non-overlapping patches and stitching predictions back together — this detail is preserved without ballooning the memory demands like you would if you tried to upsample the entire model.

10.1.4 No Post-Processing

Right now, the pipeline outputs raw argmax predictions without any spatial post-processing. Steps like connected-component analysis — useful for removing spurious small predictions — or conditional random fields for boundary refinement are standard in clinical workflows. Adding them would improve HD95 scores on organs like the esophagus and pancreas, which occasionally produce disconnected prediction islands.

10.2 Future Directions

10.2.1 2.5D and Pseudo-3D Architectures

The most immediate extension is to move from pure 2D to a 2.5D input — stacking the target slice with its immediate axial neighbours to give the model three-slice context. This approach is memory-efficient and has been shown to recover most of the performance gap between 2D and full 3D models on abdominal CT data.

10.2.2 Semi-Supervised Learning

FLARE22 includes an additional 2,000 unlabelled CT volumes. A semi-supervised approach — using the labelled 50 volumes for supervised training and the unlabelled 2,000 for consistency regularisation or pseudo-labelling — could dramatically expand the effective training set without the cost of manual annotation.

10.2.3 Uncertainty Estimation

Right now, the model produces a hard argmax prediction with no indication of confidence. Adding Monte Carlo dropout at inference — leaving the bottleneck Dropout2d active during evaluation — would produce per-voxel uncertainty estimates at negligible cost.

Uncertainty maps are clinically useful: they flag regions where the model is unsure and a radiologist should verify the segmentation.

10.2.4 Transformer-Based Backbones

Recent architectures such as Swin-UNETR replace convolutional encoders with Swin Transformers, capturing long-range spatial dependencies that convolutions miss. While these models require more data to train from scratch, fine-tuning a publicly available Swin-UNETR checkpoint on FLARE22 data would be feasible within the same hardware constraints as this project.

10.2.5 Multi-Dataset Pretraining

Training a shared backbone on several abdominal CT datasets (e.g., KiTS23, MSD-Liver, CHAOS) before fine-tuning on FLARE22 could improve robustness to scanner variability and organ co-occurrence patterns. This multi-dataset strategy is increasingly standard in large-scale medical imaging benchmarks and represents a natural next step for this work.

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